Decision Tree

- Supervised Machine Learning Algorithm that uses a set of rules to make decisions.
- A flowchart-like structure used to make decisions or predictions.
- Non-parametric method used for classification and regression.
- The different types of decision tree algorithm are- CART, ID3 and C4.5

Why decision tree are non-parametric?

- It does not assume any parametric form for the class densities
- The tree structure is not fixed apriori, i.e tree grows during learning.
- If the decision tree is only used for classification, then it is known as Classification Tree
- If the decision tree is used for regression, it is called as Regression Tree.

Advantages:

1. Interpretability: Trees can be converted into a set of IF-THEN rules; hence preferred over more accurate but less interpretable methods
2. Able to capture non-linear relationships
3. Can handle any type of data ,
   i.e, numerical, categorical, boolean.
4. Faster in execution and efficient
5. Normalization is not required

Disadvantages:

1. As no.of records increases the complexity also increases
2. Also increases training time
3. Leads to overfitting

Framework

1. Start with a tree with a single leaf, assign this leaf a label according to a majority vote among all the labels over the training set.
2. Perform series of iterations. On each iteration, examine the effect of splitting a single leaf.
3. We define a gain measure to quantify the improvement.

Attribute Selection Methods

- ***Information Gain***
- ***Gini Index***
- **Entropy:** is the measure of uncertainty of a random variable. The higher the entropy more the information content.

$$E = -\sum_{i=1}^{n} p_i \log_2(p_i)$$

- **Information gain** is a measure of this change in entropy.

$$Gain(S, A) = Entropy(S) - \sum_{v \epsilon Values(A)} \frac{|S_v|}{|S|}.Entropy(S_v)$$

ID3 Algorithm

- It constructs a tree by selecting attributes that maximize Information Gain, which is computed using entropy.
- ID3 follows a greedy, top-down recursive strategy to split the dataset until the classification becomes pure or no attributes remain.

Working

- **Initialize the Dataset**
- **Compute Entropy of Target Attribute**
- **Calculate Information Gain for Each Attribute**
- **Select the Best Attribute for Splitting**
- **Partition the Dataset**
- **Create Child Nodes Recursively**
- **Check Stopping Conditions**
- **Assign Class Labels to Leaf Nodes**

age	income	student	Credit rating	Buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

1. Calculate the entropy for the entire dataset

$$Entropy(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.94$$

Age

1. Entropy and Information gain of age

$$Entropy(age) = \frac{5}{14}\left(-\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)\right) +$$

$$\frac{4}{14}(0) + \frac{5}{14}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right)$$

$$Gain(age) = 0.94 - 0.6935 = 0.2465$$

Income

Entropy and Information gain for Income

$$Entropy(income) = \frac{4}{14}\left(-\frac{2}{4}\log_2\frac{2}{4}-\frac{2}{4}\log_2\frac{2}{4}\right)+$$

$$\frac{6}{14}\left(-\frac{4}{6}\log_2\frac{4}{6}-\frac{2}{6}\log_2\frac{2}{6}\right)+\frac{4}{14}\left(-\frac{3}{4}\log_2\frac{3}{4}-\frac{1}{4}\log_2\frac{1}{4}\right)$$

$$Gain(income) = 0.94 - 0.9108 = 0.0292$$

Student

$$Entropy(student) = \frac{7}{14}\left(-\frac{6}{7}log_2\frac{6}{7}-\frac{1}{7}log_2\frac{1}{7}\right)+$$

$$\frac{7}{14}\left(-\frac{3}{7}log_2\frac{3}{7}-\frac{4}{7}log_2\frac{4}{7}\right)$$

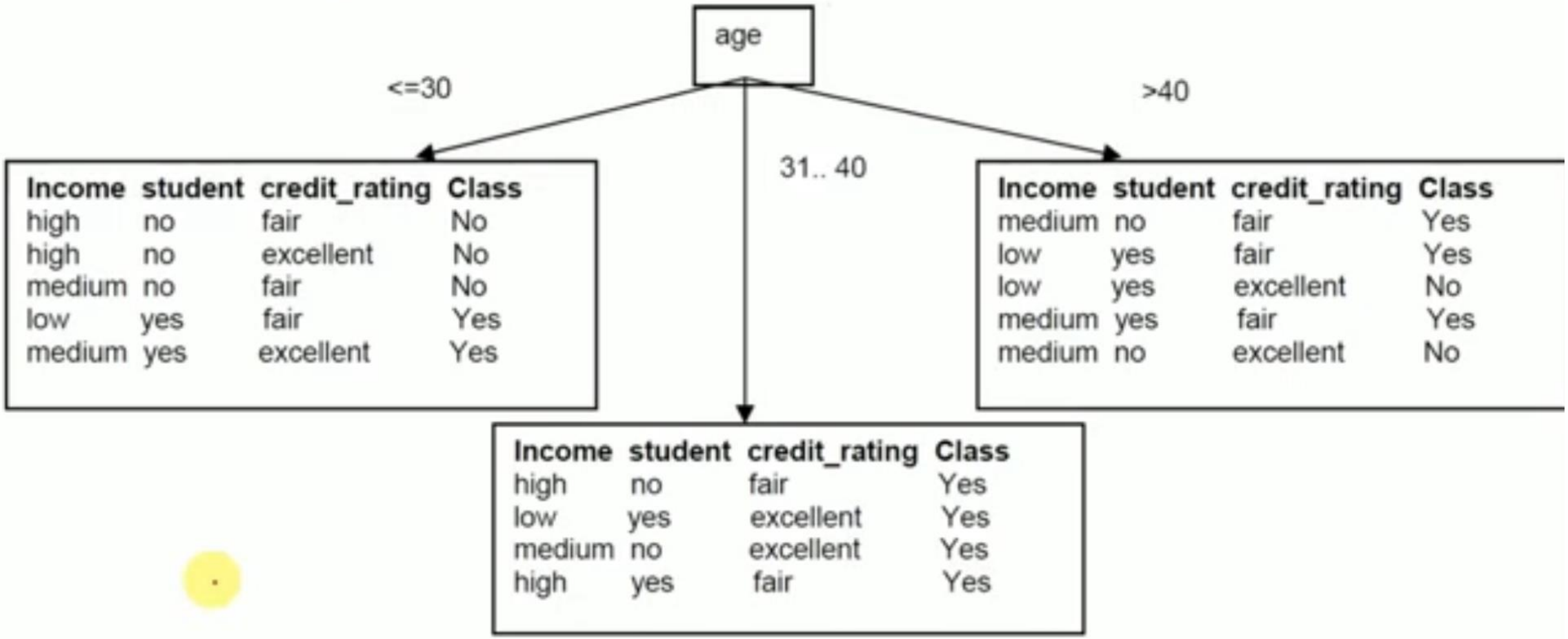
$$Gain\,(student) = 0.94 - 0.7884 = 0.1516$$

Credit rating

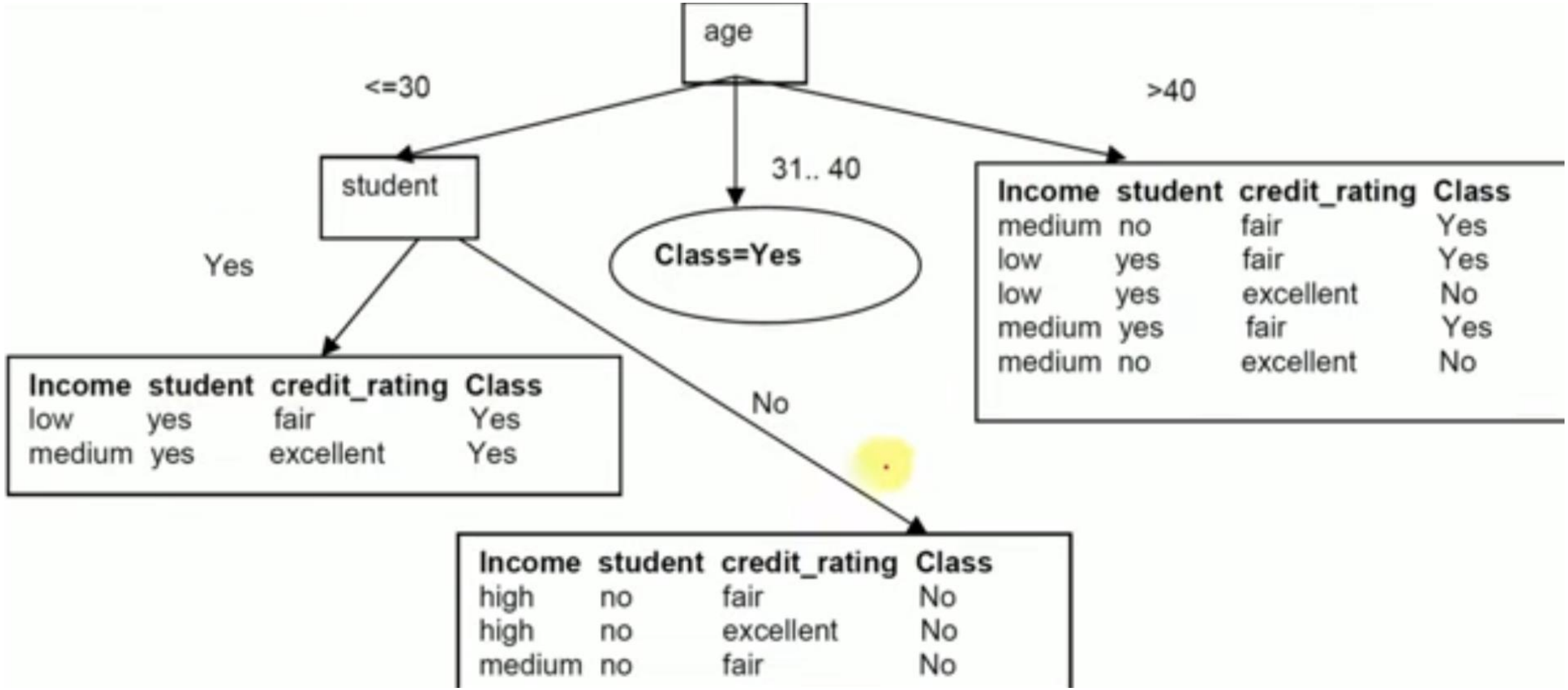
$$Entropy(credit_{rating}) = \frac{8}{14}\left(-\frac{6}{8}log_2\frac{6}{8}-\frac{2}{8}log_2\frac{2}{8}\right)+$$

$$\frac{6}{14}\left(-\frac{3}{6}log_2\frac{3}{6}-\frac{3}{6}log_2\frac{3}{6}\right)$$

$$Gain(credit_rating) = 0.94 - 0.8920$$


age
<=30
31.. 40
>40
Income student credit_rating Class
high no fair No
high no excellent No
medium no fair No
low yes fair Yes
medium yes excellent Yes
Income student credit_rating Class
high no fair Yes
low yes excellent Yes
medium no excellent Yes
high yes fair Yes
Income student credit_rating Class
medium no fair Yes
low yes fair Yes
low yes excellent No
medium yes fair Yes
medium no excellent No

Income	student	credit_rating	Class
high	no	fair	No
high	no	excellent	No
medium	no	fair	No
low	yes	fair	Yes
medium	yes	excellent	Yes


age
<=30
31.. 40
>40
student
Yes
Class=Yes
No
Income student credit_rating Class
low yes fair Yes
medium yes excellent Yes
Income student credit_rating Class
high no fair No
high no excellent No
medium no fair No
Income student credit_rating Class
medium no fair Yes
low yes fair Yes
low yes excellent No
medium yes fair Yes
medium no excellent No

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No